Exploring Aladin Sky Atlas through sound with SoniAladin

SoniAladin Installation guidelines

Download the full package from:

https://drive.google.com/drive/folders/1eKwrJuPZZoxo2iVf9166k-EStZnv6OKF?usp=sharing

Alternative download from Github: https://github.com/AuditoryVO/ICAD2026_Workshop

- **For mac users:**

1. SoniAladin runs in Python. Python is included in every macOS distribution package.
   a. Open *Terminal* and check your current python version. Write the following command and press enter:
      `$ python3 -V (or python3 --version)`
   b. If it is the first time you run python on your computer, you'll need to pre-install Xcode developer tools. Follow the installation wizard. Once Xcode is installed, check your Python version again.

2. SoniAladin requires several libraries which are listed in the provided *requirements.txt* file. To install them you will need first to install *pip*, the python standard package manager.
   a. On Terminal write the following command and press enter:
      `$ python3 -m ensurepip --upgrade`

   We will also create a virtual environment to run the project, preventing package version conflicts and keeping your Python installation clean.
   a. Install *venv*: `$ pip3 install virtualenv`
   b. Create the virtual environment: : `python3 -m virtualenv SoniAladin`
   c. Activate it: `$ source SoniAladin/bin/activate`
   d. Check the Python version in the new environment: `$ which python`
   e. Not now, but at the end of the session you can deactivate the virtual environment running: `$ deactivate`

3. Now you can install the dependencies in the virtual environment running:
   `$ pip3 install -r requirements.txt`

   Important note: if Terminal returns an error related to the versions of the libraries you can delete the version reference in the requirements.txt file. For instance, replace "`numpy==2.2.6`" just by "`numpy`", save the file and run the command above again in Terminal.

4. Once all dependencies are installed you'll use the Driver IAC to send MIDI messages to the sound engine. To activate it:
    a. Open the *Audio and MIDI configuration* of your computer.
    b. Open the MIDI Studio at Window/Show MIDI Studio.
    c. Double click over Driver IAC.
    d. Activate the device selecting "*Device is online*".

5. We're almost ready! We will use GarageBand as your default sound engine.
    a. Open GarageBand
    b. Create a new file doing: *File/New/Empty Project*, then *MIDI Software Instrument*. The "Classic Electric Piano" is loaded by default. We will use this instrument to start, later you will have time to explore different sounds or different DAWs and synthesizers.

6. We will explore the Universe using Aladin Lite
    a. Open your web browser and go to: https://aladin.cds.unistra.fr/AladinLite/
    b. In the target field (automatically selected when Aladin lite is opened), write "Polaris" to go to the North Star.

7. Finally, run SoniAladin:
    a. On Terminal, navigate to the folder where you downloaded SoniAladin
    b. Run SoniAladin with:
    ```
    $ python3 SoniAladin.py
    ```
    c. Move to Aladin lite window (Command+Tab)
    d. Say "Music" to start the sonification.
    Important note: As SoniAladin uses screen captures and speech recognition to generate the sonifications, you'll need to allow access from Terminal to your microphone, keyboard and screen recording.
    e. Once all accesses are allowed you can control SoniAladin by voice and with the keyboard while navigating Aladin Lite Atlas.

8. You can modify SoniAladin code to customize your sonifications. You'll need a code editor such as Visual Studio (https://code.visualstudio.com/Download).

- **SoniAladin control (all OS):**

Say 'music' or press 'right arrow' to start the sonification. Say 'stop' or press 'left arrow' to stop the sonification. Say 'exit' or press'Esc' to finish.

- **For windows users:**

1. SoniAladin runs in Python. Install Python in your computer from: https://www.python.org/downloads/windows/. Current version at the moment is 3.13.12. Python13.14 is not compatible with rtmidi dependencies. Python 3.12 is recommended.

2. SoniAladin requires several libraries which are listed in the provided *requirements.txt* file. To install them you will need first to install *pip*, the python standard package manager.
    a. On Terminal write the following command and press enter:
    `C:> python -m ensurepip --upgrade`

    We will also create a virtual environment to run the project. It prevents package version conflicts and keeps your Python installation clean. Navigate to the folder where you have the workshop files using “cd”.
    f. Install *venv* : `C:> python –m pip install virtualenv`
    g. Create the virtual environment: :
    `C:> python -m venv SoniAladin_venv`
    If you have several python installations and want to use a specific one (3.12):
    `C:> py -3.12 -m venv SoniAladin_venv`
    h. Activate the virtual environment:
    `PowerShell => C:> .\SoniAladin_venv\Scripts\Activate.ps1`
    `Command Prompt => C:> source SoniAladin_venv/bin/activate`
    i. Check the Python version in the new environment: `C:> python --version`
    j. Not now, but at the end of the session you can deactivate the virtual environment running: `C:> deactivate`

3. Now you can install the dependencies in your virtual environment running:
    `C:> python -m pip install -r requirements.txt`

    Important note: if Terminal returns an error related to the versions of the libraries you can delete the version reference in the requirements.txt file. For instance, replace “`numpy==2.2.6`“ just by “`numpy`”, save the file and run the command above again in Terminal.

4. SoniAladin automatically picks Microsoft GS Wavetable Synth, and plays directly. GS Wavetable is fully built in and reachable as a named MIDI output port, so Python MIDI output sounds with no extra configuration. We will use this instrument to start, later you will have time to explore different GM instruments or different DAWs and synthesizers.

    To change the GM Wavetable instruments, you can easily modify SoniAladin code. You’ll need a code editor such as Visual Studio (https://code.visualstudio.com/Download) General MIDI instruments are assigned to programs (0-127). The following example changes channel 1 from piano (0) to harp (46):

    ```
    channel = 1
    program_number = 46
    ```

```
status = 0xC0 + (channel - 1)
midi_out.send_message([status, program_number])
```

Piano	Bass	Reed	Synth Effects
0 Acoustic Grand Piano	32 Acoustic Bass	64 Soprano Sax	96 SFX Rain
1 Bright Acoustic Piano	33 Electric Fingered Bass	65 Alto Sax	97 SFX Soundtrack
2 Electric grand Piano	34 Electric Picked Bass	66 Tenor Sax	98 SFX Crystal
3 Honky Tonk Piano	35 Fretless Bass	67 Baritone Sax	99 SFX Atmosphere
4 Electric Piano 1	36 Slap Bass 1	68 Oboe	100 SFX Brightness
5 Electric Piano 2	37 Slap Bass 2	69 English Horn	101 SFX Goblins
6 Harpsichord	38 Syn Bass 1	70 Bassoon	102 SFX Echoes
7 Clavinet	39 Syn Bass 2	71 Clarinet	103 SFX Sci-Fi
Chromatic Percussion	**Strings/Orchestra**	**Pipe**	**Ethnic**
8 Celesta	40 Violin	72 Piccolo	104 Sitar
9 Glockenspiel	41 Viola	73 Flute	105 Banjo
10 Music Box	42 Cello	74 Recorder	106 Shamisen
11 Vibraphone	43 Contrabass	75 Pan Flute	107 Koto
12 Marimba	44 Tremolo Strings	76 Bottle Blow	108 Kalimba
13 Xylophone	45 Pizzicato Strings	77 Shakuhachi	109 Bag Pipe
14 Tubular bells	46 Orchestral Harp	78 Whistle	110 Fiddle
15 Dulcimer	47 Timpani	79 Ocarina	111 Shanai
Organ	**Ensemble**	**Synth Lead**	**Percussive**
16 Drawbar Organ	48 String Ensemble 1	80 Syn Square Wave	112 Tinkle Bell
17 Percussive Organ	49 String Ensemble 2	81 Syn Sawtooth Wave	113 Agogo
18 Rock Organ	50 Syn Strings 1	82 Syn Calliope	114 Steel Drums
19 Church Organ	51 Syn Strings 2	83 Syn Chiff	115 Woodblock
20 Reed Organ	52 Choir Aahs	84 Syn Charang	116 Taiko Drum
21 Accordion	53 Voice Oohs	85 Syn Voice	117 Melodic Tom
22 Harmonica	54 Syn Choir	86 Syn Fifths Sawtooth	118 Syn Drum
23 Tango Accordion	55 Orchestral Hit	87 Syn Brass & Lead	119 Reverse Cymbal
Guitar	**Brass**	**Synth Pad**	**Sound Effects**
24 Nylon Acoustic	56 Trumpet	88 New Age Syn Pad	120 Guitar Fret Noise
25 Steel Acoustic	57 Trombone	89 Warm Syn Pad	121 Breath Noise
26 Jazz Electric	58 Tuba	90 Polysynth Syn Pad	122 Seashore
27 Clean Electric	59 Muted Trumpet	91 Choir Syn Pad	123 Bird Tweet
28 Muted Electric	60 French Horn	92 Bowed Syn Pad	124 Telephone Ring
29 Overdrive	61 Brass Section	93 Metal Syn Pad	125 Helicopter
30 Distorted	62 Syn Brass 1	94 Halo Syn Pad	126 Applause
31 Harmonics	63 Syn Brass 2	95 Sweep Syn Pad	127 Gun Shot

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6. Finally, run SoniAladin:
   a. On Terminal, navigate to the folder where you downloaded SoniAladin
   b. Run SoniAladin with:

      ```
      C:> py SoniAladin.py
      ```
   c. Move to Aladin lite window (Command+Tab)
   d. Say "Music" to start the sonification.
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- **Resources for the exploration:**

  IAU constellation maps
  https://iauarchive.eso.org/public/themes/constellations/

Constellation	Abbreviation	English name
Ursa Major	UMa	The great bear
Ursa Minor	UMi	The little bear
Orion	Ori	The hunter
Cassiopeia	Cas	The Queen
Saggitarius	Sgr	The archer
Leo	Leo	The Lion
Taurus	Tau	The bull
Gemini	Gem	The twins
Andromeda	And	The princess
Pegasus	Peg	The winged horse

IAU star names
https://iauarchive.eso.org/public/themes/naming_stars/

IAU exoplanet names
https://iauarchive.eso.org/public/themes/naming_exoplanets/

Wiki list of galaxies
https://simple.wikipedia.org/wiki/List_of_galaxies